

1. Update Kali Linux

1. Open a terminal in Kali.
2. Run:

```
bash
sudo apt update && sudo apt -y full-upgrade
```

This command will update the kali linux packages and its environment to the latest version.

2. Install Apache, MariaDB, and PHP

1. In the terminal, install the required services and PHP modules:

```
bash
sudo apt install apache2 mariadb-server php php-mysqli php-gd libapache2-mod-php -y
```

Apache provides the web server; MariaDB provides the database; PHP and modules are required for DVWA.

2. Enable and start services:

```
bash
sudo systemctl enable apache2 mariadb
sudo systemctl start apache2 mariadb
```

This ensures the web server and database start now and on boot.

3. Download DVWA into the Web Root

1. Change to the Apache web root directory:

```
bash
cd /var/www/html
```

DVWA must reside under the web root to be served by Apache.

Go to the web root directory where DVWA will be downloaded so that it can run Apache.

2. Install Git if needed, then clone DVWA:

```
bash
sudo apt install git -y
sudo git clone https://github.com/digininja/DVWA.git
```

This downloads the DVWA source code into web root directory -
`/var/www/html/DVWA`.

4. Set DVWA Directory Permissions

1. Verify the DVWA directory exists:

```
bash
ls
```

Confirm a DVWA is downloaded and located in the web root directory

2. Set broad permissions so the web server can read and write files for lab use:

```
bash
sudo chmod -R 777 DVWA
```

This command gives DVWA full read, write and execute permissions (777).

5. Prepare the DVWA Configuration File

1. Change into the DVWA configuration directory:

```
bash
cd DVWA/config
```

2. Copy the default configuration template:

```
bash
sudo cp config.inc.php.dist config.inc.php
```

DVWA uses `config.inc.php` as the active configuration file.

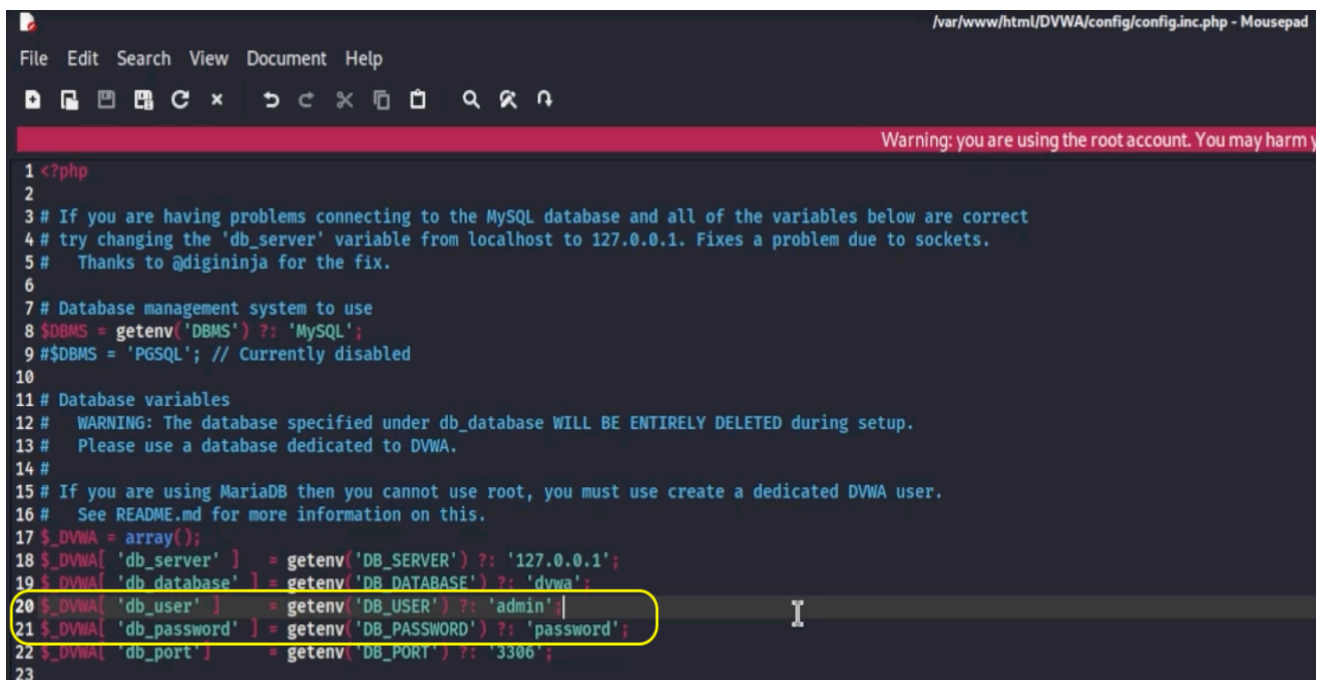
3. Open `config.inc.php` in a text editor:

```
bash
sudo mousepad config.inc.php
```

4. Set database credentials in the file, for example:

- Database user: **admin**
- Database password: **password**

Update the corresponding lines in `config.inc.php`, then save and close.



```
1 <?php
2
3 # If you are having problems connecting to the MySQL database and all of the variables below are correct
4 # try changing the 'db_server' variable from localhost to 127.0.0.1. Fixes a problem due to sockets.
5 # Thanks to @digininja for the fix.
6
7 # Database management system to use
8 $DBMS = getenv('DBMS') ?: 'MySQL';
9 # $DBMS = 'PGSQL'; // Currently disabled
10
11 # Database variables
12 # WARNING: The database specified under db_database WILL BE ENTIRELY DELETED during setup.
13 # Please use a database dedicated to DVWA.
14 #
15 # If you are using MariaDB then you cannot use root, you must use create a dedicated DVWA user.
16 # See README.md for more information on this.
17 $_DVWA = array();
18 $_DVWA['db_server'] = getenv('DB_SERVER') ?: '127.0.0.1';
19 $_DVWA['db_database'] = getenv('DB_DATABASE') ?: 'dvwa';
20 $_DVWA['db_user'] = getenv('DB_USER') ?: 'admin';
21 $_DVWA['db_password'] = getenv('DB_PASSWORD') ?: 'password';
22 $_DVWA['db_port'] = getenv('DB_PORT') ?: '3306';
23
```

6. Create the DVWA Database and User

1. Start a MariaDB shell as root:

```
bash
sudo mysql -u root -p
```

Leave the password blank and press Enter if no root password is configured.

2. Create the DVWA database:

```
sql
CREATE DATABASE dvwa;
```

3. Create a dedicated DVWA user and grant privileges, matching the credentials you set in config.inc.php (example: admin / password):

```
sql
CREATE USER 'admin'@'127.0.0.1' IDENTIFIED BY 'password';
GRANT ALL PRIVILEGES ON dvwa.* TO 'admin'@'127.0.0.1';
FLUSH PRIVILEGES;
EXIT;
```

```
root@ms: /var/www/html/DVWA/config
Session Actions Edit View Help

(root@ms)-[/var/www/html/DVWA/config]
mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 32
Server version: 11.8.3-MariaDB-1+b1 from Debian -- Please help get to 10k stars at https://github.com/MariaDB/Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> create database dvwa;
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]> create user 'admin'@'127.0.0.1' identified by 'password';
Query OK, 0 rows affected (0.008 sec)

MariaDB [(none)]> grant all privileges on dvwa.* to 'admin'@'127.0.0.1';
Query OK, 0 rows affected (0.006 sec)

MariaDB [(none)]> exit
Bye
```

This allows DVWA to connect to the `dvwa` database from localhost using the specified user.

7. Adjust PHP Configuration (`allow_url_include`)

1. Locate the PHP configuration file under Apache's configuration directory (for example, `/etc/php/8.x/apache2/php.ini`, version may differ).
2. Change to the directory and edit `php.ini`:

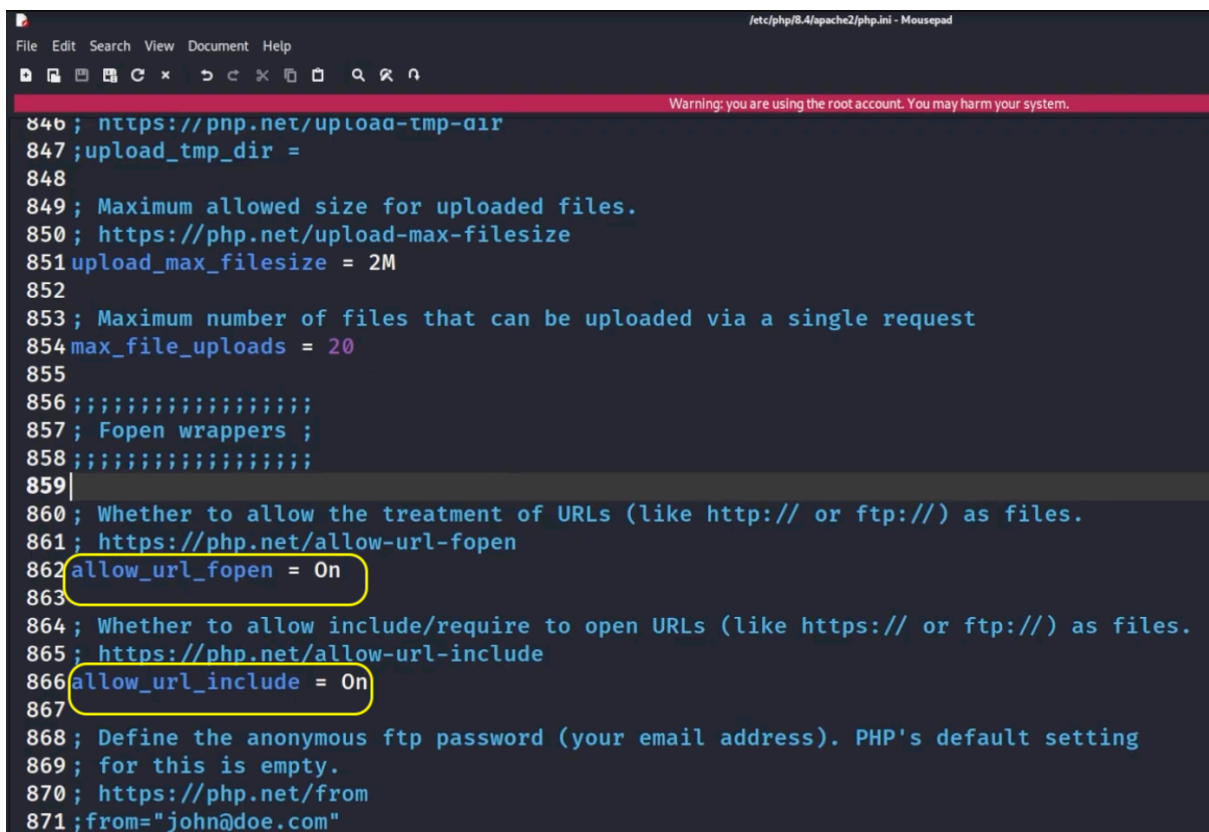
```
bash
cd /etc/php/8.X/apache2
sudo mousepad php.ini
```

Replace `8.x` with the latest version of PHP.

3. Use the editor's search function to find `allow_url_fopen` and `allow_url_include`.
4. Ensure both directives are set to `On`:

```
text
allow_url_fopen = On
allow_url_include = On
```

DVWA requires `allow_url_include` for some vulnerabilities to function; the video explicitly enables these flags.



```
File Edit Search View Document Help
Warning: you are using the root account. You may harm your system.
846 ; https://pnp.net/upload-tmp-dir
847 ;upload_tmp_dir =
848
849 ; Maximum allowed size for uploaded files.
850 ; https://php.net/upload-max-filesize
851 upload_max_filesize = 2M
852
853 ; Maximum number of files that can be uploaded via a single request
854 max_file_uploads = 20
855
856 ;;;;;;;;;;;;;;;;;;
857 ; Fopen wrappers ;
858 ;;;;;;;;;;;;;;;;;;
859
860 ; Whether to allow the treatment of URLs (like http:// or ftp://) as files.
861 ; https://php.net/allow-url-fopen
862 allow_url_fopen = On
863
864 ; Whether to allow include/require to open URLs (like https:// or ftp://) as files.
865 ; https://php.net/allow-url-include
866 allow_url_include = On
867
868 ; Define the anonymous ftp password (your email address). PHP's default setting
869 ; for this is empty.
870 ; https://php.net/from
871 ;from="john@doe.com"
```

5. Save changes and close the editor.

8. Restart Apache

1. Restart Apache to apply the PHP configuration changes:

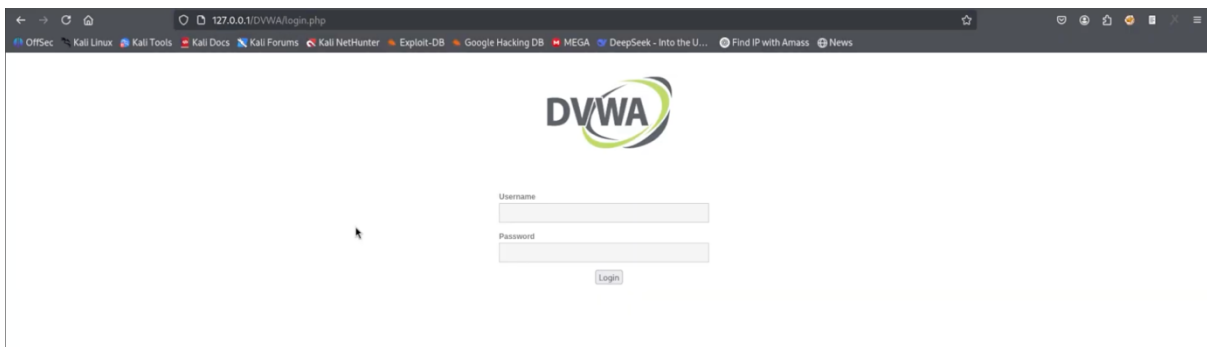
```
bash
sudo systemctl restart apache2
```

9. Access DVWA in the Browser

1. Open a web browser inside Kali (for example, Firefox).
2. Navigate to:

```
text
http://127.0.0.1/DVWA
```

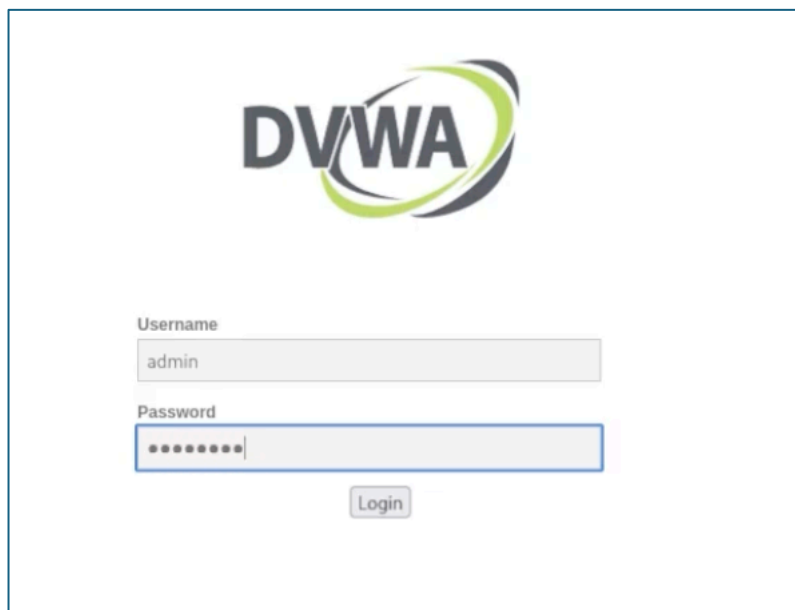
This should load the DVWA login/setup page.



3. If you see a warning about missing database tables, click the **Setup** or **Create / Reset Database** button in the DVWA interface to initialise the schema.

10. Log In and Verify

1. On the DVWA login page, use the default credentials:
 - Username: admin
 - Password: password



2. After logging in, confirm that DVWA loads the main interface and that modules (such as SQL Injection or XSS) are available.
3. Optionally set **DVWA Security** to **Low** to begin testing, then raise the level as needed for your experiments.

127.0.0.1/DVWA/index.php

Kali DocsKali ForumsKali NetHunterExploit-DBGoogle Hacking DBMEGADeepSeek - Into the U...Find IP with AmassNews



Home

Instructions

Setup / Reset DB

Brute Force

Command Injection

CSRF

File Inclusion

File Upload

Insecure CAPTCHA

SQL Injection

SQL Injection (Blind)

Weak Session IDs

XSS (DOM)

XSS (Reflected)

XSS (Stored)

CSP Bypass

JavaScript Attacks

Authorisation Bypass

Open HTTP Redirect

Cryptography

API

DVWA Security

PHP Info

About

Logout

Welcome to Damn Vulnerable Web Application!

Damn Vulnerable Web Application (DVWA) is a PHP/MySQL web application that is damn vulnerable. Its main goal is to be an aid for security professionals to test their skills and tools in a legal environment, help web developers better understand the processes of securing web applications and to aid both students & teachers to learn about web application security in a controlled class room environment.

The aim of DVWA is to **practice some of the most common web vulnerabilities**, with various levels of difficulty, with a simple straightforward interface.

General Instructions

It is up to the user how they approach DVWA. Either by working through every module at a fixed level, or selecting any module and working up to reach the highest level they can before moving onto the next one. There is not a fixed object to complete a module; however users should feel that they have successfully exploited the system as best as they possible could by using that particular vulnerability.

Please note, there are **both documented and undocumented vulnerabilities** with this software. This is intentional. You are encouraged to try and discover as many issues as possible.

There is a help button at the bottom of each page, which allows you to view hints & tips for that vulnerability. There are also additional links for further background reading, which relates to that security issue.

WARNING!

Damn Vulnerable Web Application is damn vulnerable! **Do not upload it to your hosting provider's public html folder or any Internet facing servers**, as they will be compromised. It is recommend using a virtual machine (such as [VirtualBox](#) or [VMware](#)), which is set to NAT networking mode. Inside a guest machine, you can download and install [XAMPP](#) for the web server and database.

Disclaimer

We do not take responsibility for the way in which any one uses this application (DVWA). We have made the purposes of the application clear and it should not be used maliciously. We have given warnings and taken measures to prevent users from installing DVWA on to live web servers. If your web server is compromised via an installation of DVWA it is not our responsibility it is the responsibility of the person/s who uploaded and installed it.

More Training Resources

DVWA aims to cover the most commonly seen vulnerabilities found in today's web applications. However there are plenty of other issues with web applications. Should you wish to explore any additional attack vectors, or want more difficult challenges, you may wish to look into the following other projects:

- [Mutillidae](#)
- [OWASP Vulnerable Web Applications Directory](#)